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MIL-H-28523B(NAVY)
9 JANUARY 1990
SUPERSEDING
MIL-H-28523A(YD)
2 March 1981

MILITARY SPECIFICATION

HOSE ASSEMBLY, RUBBER, HYDRAULIC AND PNEUMATIC, JETTING (400 PSI WORKING PRESSURE)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hose assemblies for hydraulic and pneumatic jetting in pile driving, rock drilling, and air compressor service.

1.2 Classification. Hose covered by this specification shall be of one type.

1.2.1 Size. Hose covered by this specification shall be the inside diameter sizes listed in 3.7.1 as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

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2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing): and Fittings, Nozzles, and Strainers.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

ASTM

ASTM D 380 - Testing Rubber Hose.
ASTM D 412 - Tension Testing of Vulcanized Rubber.
ASTM D 413 - Rubber Property - Adhesion to Flexible Substrate.
ASTM D 471 - Rubber Property - Effect of Liquids.
ASTM D 518 - Rubber Deterioration - Surface Cracking.
ASTM D 573 - Rubber Deterioration in Air Oven.
ASTM D 1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimens).
ASTM D 1193 - Reagent Water.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a complete hose assembly shall be subjected to first article inspection (see 4.2.1, 6.2, and 6.5).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.2.1 Inner tube. The rubber compound used for the inner tube shall be compounded from synthetic rubber to resist water and oil. The inner tube shall have a smooth bore, free of pitting, dirt, and mandrel lubricants, and shall have a minimum thickness of not less than 0.093 inches (in.).

3.2.2 Reinforcement. The reinforcement shall consist of sufficient plies of wrapped duck, spiral or braided yarns of cotton or synthetic fibers, or wire, or a combination of these necessary to meet the requirements of this specification and be suitable for the purpose intended. Where used, the layers of duck will be of cotton or synthetic fibers free from mechanical defects and applied on a bias of 45 degrees (°) with edges lapped not less than 0.5 inch.

3.2.3 Cover. The cover shall consist of an oil-, ozone-, and abrasion-resistant compound made from polymerized chloroprene or an equally oil-, ozone-, and abrasion-resisting material to meet the requirements of this specification. The cover shall have a thickness of not less than 0.063 inch.

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3.3 Construction. Hose assemblies shall consist of tube, reinforcement and cover as described in 3.2.1, 3.2.2, and 3.2.3 with the reinforcement plies (fabric, yarn, or wire) suitably impregnated with rubber to achieve a continuous rubber band from the tube through the reinforcement to the outer cover. They shall be fitted with swivel connections as described in 3.4.

3.4 Hose end fittings. The ends of the hose shall be fitted with pearlitic malleable iron, interlocking type female couplings. Couplings shall be provided with a wing nut with wrenching or knock lug features for quick loosening. The wing nut shall have NH hose threads to hold the steel adapter spud against the ground seat. The spud shall have internal NPT threads on one end and external NH threads with a mating ground joint seat on the other end. A 4-bolt clamp with four fingers shall be used to hold the shank to the hose. Threads shall be in accordance with FED-STD-H28.

3.5 Physical requirements. The hose and hose assembly shall conform to the requirements specified in table I and the following paragraphs when tested as specified in Section 4.5.

TABLE I. Performance requirements.

Hydrostatic test pressures (minimum)	
Proof pressure (psi)	800
Burst pressure (psi)	1600
Tensile strength of rubber compound (before aging)(minimum)	
Tube, psi	850
Cover, psi	1000
Elongation (minimum)	
Tube (percent)	175
Cover (percent)	200
Adhesion (minimum)	
Between plies and between rubber parts (pounds-force/inch)	12

3.5.1 Resistance to water.

3.5.1.1 Swelling. The increase in the volume of the tube shall not exceed 20 percent when tested as described in 4.5.4.

3.5.2 Resistance to oil. The tensile strength of the tube and cover shall not be less than 700 pounds per square inch (psi) when tested as specified in 4.5.1.

3.5.3 Accelerated aging. The tensile strength of the tube and cover after the accelerated aging process described in 4.5.6 shall not be less than 80 percent of the tensile strength indicated before the aging process, and ultimate elongation shall be not less than 50 percent of the values specified in table I.

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3.5.4 Ozone resistance. The cover compound of the hose shall show no visible cracking under 2X magnification after exposure to ozone as specified in 4.5.8.

3.6 Identification marking. Each length of hose shall be marked on the cover in two places, approximately 4 feet (ft) from the ends. The marking shall be in letters not less than 0.125 inch high and shall include: (a) the manufacturer's name or trademark, (b) the maximum working pressure (e.g., working pressure 400 psi), and (c) the quarter of the calendar year and the year of manufacture.

3.7 Dimensions.

3.7.1 Hose diameter. The inside diameter of the hose shall be 2 inches or 3 inches (±0.062), as specified (see 6.2).

3.7.2 Length. Unless otherwise specified (see 6.2), the hose length shall be 50 feet. The length of the hose shall be measured from the ends of the hose, not from the ends of the couplings.

3.7.2.1 Tolerance. A tolerance of ±1 percent (%) shall be permitted in the length of the hose.

3.8 Workmanship. The finished hose assembly shall be free of any defect that may affect its appearance and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one sample hose length when a first article is required (see 3.1, 6.2, and 6.5). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All hose and hose assemblies of the same type and size offered for delivery at one time shall be considered a lot for the purpose of inspection.

4.3.1 Sampling for examination. (See 6.7.1.)

4.3.2 Sampling for tests. (See 6.7.2.)

4.4 Examination. Each of the sample hose lengths selected shall be visually and dimensionally examined to determine conformance with all the requirements of this specification not involving tests. The classification of defects in FED-STD-162 shall be used to determine and evaluate defects through visual inspection.

4.5 Tests. Sample hose sections selected shall be subjected to the applicable test specified in table II and 4.5.1 through 4.5.8.

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TABLE II. Test method references.

	Requirement paragraph	ASTM Test Method Standard	Reference paragraph
Hose size (inside diameter)	3.7.1	ASTM D 380	
Hose length	3.7.2	ASTM D 380	
Tube and cover thickness	3.2.1 and 3.2.3	ASTM D 380	
Proof pressure	3.5 and table I	ASTM D 380	4.5.1
Burst pressure	3.5 and table I	ASTM D 380	4.5.2
Tensile strength and elongation	3.5 and table I	ASTM D 412	4.5.3
Swelling	3.5.1.1	ASTM D 471	4.5.4
Accelerated aging	3.5.3	ASTM D 573	4.5.5
Adhesion	3.5 and table I	ASTM D 413	4.5.6
Oil immersion	3.5.2	ASTM D 471	4.5.7
Ozone	3.5.4	ASTM D 1149	4.5.8

4.5.1 Hydrostatic proof test, coupled assemblies. Samples selected after couplings have been reattached shall be subjected to the hydrostatic proof test pressure specified in table I in accordance with ASTM D 380. Water shall be used as the test media. Leakage or other evidence of defects shall be cause for rejection.

4.5.2 Burst test. Hose samples selected shall be subjected to the hydrostatic straight bursting test of ASTM D 380. Water shall be used at the test media. Any suitable coupling other than those specified in 3.4 may be used for this test. Failure of the hose at a pressure below the minimum burst pressure specified in table I shall be cause for rejection of the lot.

4.5.3 Initial tensile properties. Initial tensile strength and ultimate elongation shall be determined by the procedure given in ASTM D 412, using specimens cut with die C of ASTM D 412.

4.5.4 Volume change after immersion. The change in volume of samples of the tube, after immersion for 22 $\pm$ 1/4 hours at a temperature of 73 $\pm$ 4 degrees Fahrenheit (°F) in type IV reagent water of ASTM D 1193 shall be determined in accordance with ASTM D 471. The change in volume shall be reported as a percent of the original volume.

4.5.5 Accelerated aging test. Hose samples selected shall be subjected to accelerated aging in accordance with ASTM D 573, except that the time of aging shall be 70 $\pm$ 1/2 hours at 212°F $\pm$ 4°F. After aging, the tensile strength shall be not less than 80 percent of the values specified in table I, and the ultimate elongation shall be not less than 50 percent of the values specified in table I. The die for cutting the specimen shall be C of ASTM D 412.

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4.5.6 Adhesion. For hose assemblies selected prepare ring specimens as described in ASTM D 380. The adhesion shall be determined in accordance with the machine method of ASTM D 413. The rate of travel of the power-actuated grip shall be 1 inch per minute. Nonconformance to the requirements of table I shall constitute failure of this test.

4.5.7 Oil immersion. The tensile strength of the tube and cover, after immersion in oil No. 1, shall be determined in accordance with ASTM D 471, using specimens cut with die C of ASTM D 412. Tensile strength shall be based on the original cross-sectional area for the specimen.

4.5.8 Ozone resistance test. Specimens of the cover, prepared as described in Procedure A of ASTM D 518, shall be mounted in a 20 percent elongated position and tested in accordance with ASTM D 1149. After conditioning for 24 hours in an ozone-free atmosphere, the mounted specimen shall be exposed for 72 hours at 104°F ±2°F to an atmosphere containing 50 parts per hundred million of ozone. Visible cracking under 2X magnification at the end of such test shall be cause for rejection.

4.6 Packaging inspection. The preservation-packaging, packing, and marking of the hose shall be inspected to verify conformance to the requirements of Section 5 of this specification.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-P-775 with the level of preservation and the level of packing as specified (see 6.2). When specified (see 6.2), hoses shall be palletized.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose covered by this specification is intended for pile driving, grading, surface stripping, hydraulic mining, air compressor use, and other applications requiring high pressure water service and air service.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Size of hose required (see 3.7.1).
d. If hose length shall be other than 50 feet (see 3.7.2).
e. When first article is required for inspection and approval (see 3.1, 4.2.1, and 6.5).
f. Level of preservation and level of packing required (see 5.1).

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6.3 Recommended working pressure. Recommended working pressure for the hose covered by this specification is 400 psi.

6.4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.5 First article. When a first article inspection is required (see 3.1 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one hose or hose assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.6 Definitive specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the requirement options available in this specification as follows:

Hose assembly.

Definitive military specification part number	M28523	X	XX	XX
Military specification number				
Hose size code letter (see 6.6.1)				
Length in feet				
Length in inches				

6.6.1 Hose size. Hose size is designated by one code letter (see table III).

TABLE III. Hose size code letter.

Hose size code letter	A	B
Hose inside diameter (inches)	2	3

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6.7 Sampling procedures.

6.7.1 Sampling for examination. Examination of the hose shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent (%) defective for major defects and 4.0 percent defective for minor defects (see 4.3.1).

6.7.2 Sampling for tests. Tests of the hose shall be based on inspection level S-3 with an AQL of 4.0 percent defective. A representative sample 3 feet in length shall be cut from lengths selected from each lot. The contractor shall reattach couplings to the remaining length from which the 3 foot (ft) length was cut. After the couplings have been reattached, the remaining length of the hose shall be used to conduct the hydrostatic proof pressure test specified in 4.5.1. When the remaining length has passed the hydrostatic proof test, it shall be accepted as a full length of hose. One-half (18 inches) of the 3-foot sample shall be used for the burst test; the remainder shall be used for the other tests listed in table II (see 4.3.2).

6.8 Subject term (key word) listing.

Air compressor
Air service
Pile driving
Rock drilling
Water service
High pressure

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

Review Activity:
DLA - CS

User Activity:
NAVY - SH

Preparing Activity:
Navy - YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-28523B (NAVY)	2. DOCUMENT DATE (YYMMDD) 9 January 1990.

3. DOCUMENT TITLE
HOSE ASSEMBLY, RUBBER, HYDRAULIC AND PNEUMATIC, JETTING (400 PSI WORKING PRESSURE)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DR. HOWARD A. PORTE, Project Manager Facilities Criteria Development Division	b. TELEPHONE (*Include Area Code*) (1) Commercial: (805) 982-5801 X439 (2) AUTOVON: 551-5801 x439
c. ADDRESS (*Include Zip Code*) Naval Construction Battalion Center Code 1563L Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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